

Development of a GPU-Accelerated Method of Moments Solver for EM Analysis of Metasurfaces

Abstract—Periodic metasurfaces printed on grounded dielectric substrates are commonly used as radar-absorbing surfaces and are typically designed using repeated evaluations of the reflection coefficient S_{11} as a function of frequency. Although full-wave finite-element method (FEM) simulations with periodic boundary conditions are accurate, they are too slow for the extensive parameter sweeps required to build datasets for data-driven design. This paper presents a spectral-domain method of moments (MoM) solver in which the substrate and ground plane are incorporated into closed-form TE/TM Floquet Green’s functions. The MoM operator is applied with fast Fourier transforms, so the dense impedance matrix is never explicitly formed, and the resulting system is solved with a restarted GMRES iteration that runs entirely on the GPU. Independent frequency points are distributed across multiple GPUs. For four representative absorber geometries, principal resonant notches agree with a commercial FEM solver to within 4.8% in frequency. A full frequency sweep on the HPC A100 GPU is up to $50\times$ faster than the same matrix-free FFT-MoM implemented in NumPy/SciPy (complex128) on the CPU.

I. INTRODUCTION

Frequency selective surfaces (FSSs) and metasurfaces are periodic arrays of patterned conductors whose reflection and transmission change with frequency [1]. When the pattern is printed on a dielectric slab with a ground plane underneath, no power is transmitted, so whatever is not reflected is absorbed. Near resonance this absorption becomes strong and appears as a notch in the reflection coefficient S_{11} . Radar-absorbing surfaces of this kind are used to reduce radar cross-section and to control electromagnetic interference [2]. The frequency, depth and width of each notch are all set by the geometry of the unit cell.

Designing an absorber therefore comes down to predicting $S_{11}(f)$ for many candidate geometries. In practice this is done with a commercial full-wave solver, for example the frequency-domain finite-element method (FEM) solver in CST Studio Suite [3], on a single unit cell with periodic boundaries and Floquet ports. Each of the four cells below takes 29–199 s in CST (Table V), which is acceptable for a few hand-tuned designs but not for the thousands of evaluations needed to train a surrogate or inverse-design model [4], [5].

For planar periodic structures, the method of moments (MoM) [6] avoids much of this cost. Only the metal pattern has to be discretized, since the substrate and ground plane can be included analytically through a spectral-domain Green’s function [7], [8]. If the pattern is meshed on a uniform grid, the MoM operator can be applied with fast Fourier transforms (FFTs) instead of being stored as a dense impedance

matrix [9]. Conventional (RWG) MoM matrix assembly and factorization have also been accelerated on GPUs [10]. These techniques have mostly been studied separately, and relatively little work has combined them in a GPU solver set up for sweeping many absorber geometries and frequencies.

In this paper we combine them in a single solver designed for throughput, that is, for evaluating many geometries and frequency points quickly rather than for solving one very large problem. Our main contributions are:

- A spectral-domain MoM formulation for impedance-sheet patterns on a grounded substrate. The slab and ground plane enter through closed-form TE/TM Floquet Green’s functions, and the operator is evaluated with FFTs, so memory grows linearly with the number of grid cells instead of with the square of the number of unknowns.
- A GPU implementation in which the restarted GMRES solve runs entirely on the device, with frequency points distributed across multiple GPUs in contiguous blocks and no communication between them.
- Validation against CST for four absorber geometries, GMRES iteration counts across the sweep, and run-time measurements on the HPC A100 GPU, showing a speedup of up to $50\times$ over the same FFT-MoM on the CPU (NumPy/SciPy, complex128).

II. FORMULATION

A. Structure and Boundary Condition

Consider a doubly periodic array with periods $a = b = 10$ mm in the plane $z = 0$, printed on a substrate of thickness h , relative permittivity ϵ_r and loss tangent $\tan \delta$, and backed by a PEC ground plane at $z = -h$ (Fig. 1). We assume an $e^{j\omega t}$ time convention and write $\epsilon_c = \epsilon_r(1 - j \tan \delta)$; k_0 and η_0 are the free-space wavenumber and impedance. The metal pattern S is a zero-thickness sheet with surface impedance Z_s , which is zero for PEC, $(1 + j)\sqrt{\omega\mu_0/2\sigma}$ for copper, and a real sheet resistance R_s for resistive films. A y -polarized plane wave of amplitude E_0 is normally incident from $z > 0$.

Without the pattern, the grounded slab acts as a short-circuited line and reflects with

$$\Gamma_{\text{bare}} = \frac{Z_{\text{in}} - \eta_0}{Z_{\text{in}} + \eta_0}, \quad Z_{\text{in}} = j \frac{\eta_0}{\sqrt{\epsilon_c}} \tan(k_0 \sqrt{\epsilon_c} h), \quad (1)$$

Fig. 1. (a) Cross-section: patterned sheet at $z = 0$, substrate ($\epsilon_r, \tan \delta, h$), PEC ground at $z = -h$, incident and reflected waves. (b) The four 10×10 mm unit-cell masks G1–G4 (253_img, 424_img, 82_img, 100_img).

Fig. 1. (a) Layered unit-cell model. (b) Unit-cell geometries G1–G4 (white = metal).

so the tangential field at $z = 0$ is $\mathbf{E}^{\text{bare}} = (1 + \Gamma_{\text{bare}})E_0\hat{\mathbf{y}}$. The current $\mathbf{J}$ on S radiates a scattered field $\mathbf{E}^s[\mathbf{J}]$ in the presence of the slab, and on the pattern

$$\mathbf{E}_t^{\text{bare}} + \mathbf{E}_t^s[\mathbf{J}] = Z_s \mathbf{J}, \quad \mathbf{r} \in S. \quad (2)$$

B. Floquet Expansion and Spectral Green's Function

Because the structure is periodic, the current and scattered field are expanded in Floquet harmonics, which at normal incidence have wavenumbers $k_{xm} = 2\pi m/a$ and $k_{yn} = 2\pi n/b$, with $k_t^2 = k_{xm}^2 + k_{yn}^2$. The tangential scattered field at $z = 0$ is

$$\mathbf{E}_t^s(x, y) = \frac{1}{ab} \sum_{m,n} \tilde{\mathbf{G}}(k_{xm}, k_{yn}) \tilde{\mathbf{J}}(k_{xm}, k_{yn}) e^{-j(k_{xm}x + k_{yn}y)}, \quad (3)$$

where $\tilde{\mathbf{J}}$ is the Fourier transform of $\mathbf{J}$ over one unit cell.

Each harmonic splits into a TM part (along $\hat{\mathbf{k}}_t$) and a TE part (along $\hat{\mathbf{z}} \times \hat{\mathbf{k}}_t$). Each part sees the air half-space in parallel with the short-circuited substrate line, so its modal Green's function is

$$G^\alpha = \frac{1}{Y_0^\alpha - jY_1^\alpha \cot(k_{zi}h)}, \quad \alpha \in \{\text{TE}, \text{TM}\}, \quad (4)$$

where $Y_i^{\text{TE}} = k_{zi}/(\omega\mu_0)$ and $Y_i^{\text{TM}} = \omega\epsilon_0\epsilon_{r,i}/k_{zi}$ are the modal admittances of air ($i = 0, \epsilon_{r,0} = 1$) and substrate ($i = 1, \epsilon_{r,1} = \epsilon_c$). The vertical wavenumbers are $k_{zi} = \sqrt{\epsilon_{r,i}k_0^2 - k_t^2}$ with $\Im\{k_{zi}\} \leq 0$ (and $\Re\{k_{zi}\} \geq 0$ when real). A superstrate (here of the same ϵ_c and h) only replaces Y_0^α by the admittance looking up into it. In Cartesian form,

$$\tilde{\mathbf{G}} = \frac{1}{k_t^2} \begin{pmatrix} k_x^2 G^{\text{TM}} + k_y^2 G^{\text{TE}} & k_x k_y (G^{\text{TM}} - G^{\text{TE}}) \\ k_x k_y (G^{\text{TM}} - G^{\text{TE}}) & k_y^2 G^{\text{TM}} + k_x^2 G^{\text{TE}} \end{pmatrix}, \quad (5)$$

which reduces to $G^{\text{TE}}\mathbf{I}$ at $k_t = 0$. Dielectric loss and the PEC ground are thus included in $\tilde{\mathbf{G}}$ rather than as extra unknowns.

C. Discretization

The unit cell is divided into an $N \times N$ grid of square cells of side $\Delta = a/N$, and the pattern is represented by its metal cells (Fig. 1b). The current is expanded in x - and y -directed rooftop basis functions $\mathbf{f}_p$ [11] on the interior edges of the metal. An x -directed rooftop is triangular in x over two adjacent cells and constant in y over one cell, so its transform is separable,

$$\tilde{f}_p^x = \Delta^2 \text{sinc}^2\left(\frac{k_x \Delta}{2}\right) \text{sinc}\left(\frac{k_y \Delta}{2}\right) e^{j(k_x x_p + k_y y_p)}, \quad (6)$$

with $\text{sinc}(u) = \sin u/u$ (and likewise for y). Galerkin testing of (2) gives the $N_u \times N_u$ system

$$\mathbf{A}\mathbf{I} = \mathbf{v}, \quad \mathbf{A} = Z_s \mathbf{T} - \mathbf{Z}, \quad Z_{pq} = \frac{1}{ab} \sum_{m,n} \tilde{\mathbf{f}}_p^* \cdot \tilde{\mathbf{G}} \tilde{\mathbf{f}}_q, \quad (7)$$

Algorithm 1 Matrix-free product $\mathbf{y} = \mathbf{A}\mathbf{I}$

Require: $\mathbf{I} = [\mathbf{I}^x; \mathbf{I}^y]$ on metal edges; kernels $\tilde{K}_{xx}, \tilde{K}_{xy}, \tilde{K}_{yy}$ on device; Z_s

- 1: $\mathbf{U}^x, \mathbf{U}^y \leftarrow \text{scatter } \mathbf{I}^x, \mathbf{I}^y \text{ onto } N \times N \text{ grids (zero off metal)}$
- 2: $\tilde{\mathbf{U}}^x, \tilde{\mathbf{U}}^y \leftarrow \text{FFT2}(\mathbf{U}^x, \mathbf{U}^y)$ ▷ batched
- 3: $\tilde{\mathbf{E}}^x \leftarrow \tilde{K}_{xx} \odot \tilde{\mathbf{U}}^x + \tilde{K}_{xy} \odot \tilde{\mathbf{U}}^y$
- 4: $\tilde{\mathbf{E}}^y \leftarrow \tilde{K}_{xy} \odot \tilde{\mathbf{U}}^x + \tilde{K}_{yy} \odot \tilde{\mathbf{U}}^y$
- 5: $\mathbf{E}^x, \mathbf{E}^y \leftarrow \text{IFFT2}(\tilde{\mathbf{E}}^x, \tilde{\mathbf{E}}^y)$
- 6: $\mathbf{y} \leftarrow Z_s \mathbf{T}\mathbf{I} - \text{gather}(\mathbf{E}^x, \mathbf{E}^y) \text{ on metal edges}$
- 7: **return** $\mathbf{y}$

TABLE I
COST PER FREQUENCY POINT

	Dense MoM + LU	FFT + GMRES
Memory	$O(N_u^2)$	$O(N^2 + mN_u)$
Setup	$O(N_u^2 N_h)$	$O(N^2)$
Solve	$O(N_u^3)$	$O(KN^2 \log N)$

N_h : Floquet terms; m : GMRES restart; K : total iterations. The FFT grid is $M = N$.

where $\mathbf{T}$ is the Gram matrix of the basis and $v_p = \langle \mathbf{f}_p, \mathbf{E}^{\text{bare}} \rangle$.

D. Reflection Coefficient

The specular reflection is carried by the $(0, 0)$ harmonic. Referenced to the sheet plane $z = 0$,

$$S_{11} = \Gamma_{\text{bare}} + \frac{T_{\text{scat}}}{E_0 ab} \hat{\mathbf{y}} \cdot \tilde{\mathbf{G}}(0, 0) \tilde{\mathbf{J}}(0, 0), \quad (8)$$

where $T_{\text{scat}} = 1$ for air above the sheet and is a scalar factor when a superstrate is present. Only co-polarized reflection is computed. A constant phase offset may remain relative to a CST Floquet port whose reference plane is not $z = 0$; it does not affect $|S_{11}|$, which is the quantity compared below.

III. GPU-ACCELERATED SOLVER

A. FFT-Based Operator Application

Because the basis functions sit on a uniform grid and the Green's function is periodic, $\mathbf{Z}$ extended to the full grid is block circulant. Its product with a vector is therefore a spectral-domain multiplication followed by restriction to the metal edges (Algorithm 1), as in the CG-FFT approach [9]. The spectral kernels $\tilde{K}_{uv} = \frac{1}{ab} \tilde{F}^{u*} \tilde{G}_{uv} \tilde{F}^v$ are computed once per frequency on an $N \times N$ FFT grid ($M = N$). Floquet harmonics with $|m|$ or $|n|$ beyond $N/2$ are not stored separately; they are folded onto this grid by aliasing, which is the discrete Green's-function sum used throughout.

Each product costs $O(N^2 \log N)$ and needs $O(N^2)$ memory, compared with $O(N_u^2)$ storage for an explicit $\mathbf{Z}$ (Table I). For $N = 120$, a fully metallized cell would give $N_u = 2N^2 = 28,800$ unknowns, so a dense complex 128×128 $\mathbf{Z}$ would occupy about 13 GB per frequency, whereas the FFT grids and kernels need about 1 MB. Patterned cells are much sparser: at $N = 120$ the four validation geometries have $N_u = 1,350$ (G1), 12,960 (G2), 1,106 (G3) and 19,752 (G4).

Fig. 2. Pipeline: geometry $\rightarrow$ mask; frequency list split across GPU 0 to GPU $P-1$; inside each GPU: build $\tilde{K} \rightarrow$ GMRES loop (Algorithm 1) $\rightarrow S_{11}(f)$; host gathers the sweep.

Fig. 2. Solver pipeline and distribution of the frequency sweep across GPUs.

B. GPU-Resident GMRES

System (7) is solved with restarted GMRES [12] (restart $m = 80$). Orthogonalization is two-pass classical Gram–Schmidt. A left Jacobi (diagonal) preconditioner is used, built from the spectral diagonal of $\tilde{G}$ plus the rooftop Gram diagonal. On the GPU the arithmetic is IEEE complex64 and the relative residual tolerance is 5×10^{-4} (complex64 GMRES stalls near 10^{-5} if a tighter 10^{-8} request is left unchanged). The CPU reference solver uses complex128 and 10^{-8} . All data stay on the GPU: the spectral kernels, the Krylov basis ($81N_u$ complex64 values, about 19 MB for the fully metallized example above), and the Hessenberg matrix, so the only host–device transfer per frequency is the final S_{11} value. FFTs use cuFFT through PyTorch [13]. Each frequency is warm-started from the previous solution (linear extrapolation of the last two solutions when the previous solve was cheap). If GMRES hits the iteration cap with a residual still above 10^{-3} , the last iterate is kept and the sweep continues; those frequencies are logged rather than aborting the run.

C. Multi-GPU Frequency Sweep

The N_f frequency points are independent, so they can be partitioned across GPUs with no inter-GPU communication; each GPU builds its own kernels and runs its own solves, and the host gathers the results (Fig. 2). The assignment is contiguous blocks (frequency indices split with `numpy.array_split`), not round-robin and not a dynamic queue. Since GMRES needs more iterations near resonance (Fig. 4), contiguous blocks can be unbalanced. On a single large GPU, several frequencies can also be batched into one GMRES launch. Workstation T1000 runs use a batch size of one; the HPC A100 times below are for the GPU assigned by the scheduler.

IV. RESULTS

A. Setup

Table II lists the solver parameters and Table III the four unit cells. The CST frequency-domain (FEM) reference uses unit-cell boundaries and Floquet ports, sampled at the same $N_f = 201$ points from 2 to 18 GHz as the MoM sweep. CST stores linear $|S_{11}|$; all dB values below are $20 \log_{10} |S_{11}|$.

B. Validation Against FEM

Fig. 3 compares $|S_{11}|$ (dB) from the proposed solver ($N = 120$) and CST. Table IV lists matched local minima; no frequency failed the GMRES residual test. The largest principal-notch error is G2 ($\Delta f = +4.76\%$); G1, G3 and G4 are within 3.3%. G4 also has a weaker first dip (FEM

TABLE II
SOLVER PARAMETERS (COMMON TO G1–G4)

Unit cell ($a \times b$)	10×10 mm
Frequency range	2–18 GHz, $N_f = 201$ ($\Delta f = 80$ MHz)
Grid	$N = 120$ ($\Delta = 83.3 \mu\text{m}$), $M = N = 120$
GMRES	restart 80; GPU rtol 5×10^{-4} (complex64); CPU rtol 10^{-8}
CPU	NumPy/SciPy, complex128
GPUs	NVIDIA T1000 8 GB; HPC NVIDIA A100
Software	Python 3.14, PyTorch 2.11 (CUDA 12.8)

4.96 GHz, MoM 6.08 GHz). Across G1–G4 the principal notches agree in frequency to within 4.8%.

C. GMRES Iterations

The spectra in Fig. 3 and Table IV use $N = 120$. Fig. 4 shows GMRES iterations on G1: they stay in the 30–80 range off resonance and rise to about 1600 at the notch, so a contiguous frequency block that contains the resonance can take much longer than a block away from it. Resistive sheets G2 and G4 stay at or below 80 iterations across the band, so the spike is a copper-notch effect rather than a property of every geometry.

D. Performance

Table V reports 201-point sweep times ($N = 120$, 2–18 GHz): the HPC A100 is $33\times$ – $50\times$ faster than the CPU FFT-MoM and $24\times$ – $253\times$ faster than CST. Scaled to a 1000-geometry dataset, the mean sweep times project to about 37 h for CST, 12 h for the CPU FFT-MoM and 17 min on the A100 (Fig. 5).

TABLE III
FOUR VALIDATION GEOMETRIES

	Type	ϵ_r	$\tan \delta$	h (mm)	Sup.	Conductor	N_u
G1	narrowband	3.45	0.0025	0.762	air	Cu SIBC	1,350
G2	moderate WB	3.54	0.0033	3.048	air	50 $\Omega/\square$	12,960
G3	multi-notch	2.94	0.0021	3.05	superstrate	Cu SIBC	1,106
G4	ultra WB	2.6	0.0013	3.175	superstrate	40 $\Omega/\square$	19,752

Cu SIBC: $\sigma = 5.8 \times 10^7$ S/m. N_u at $N = 120$.

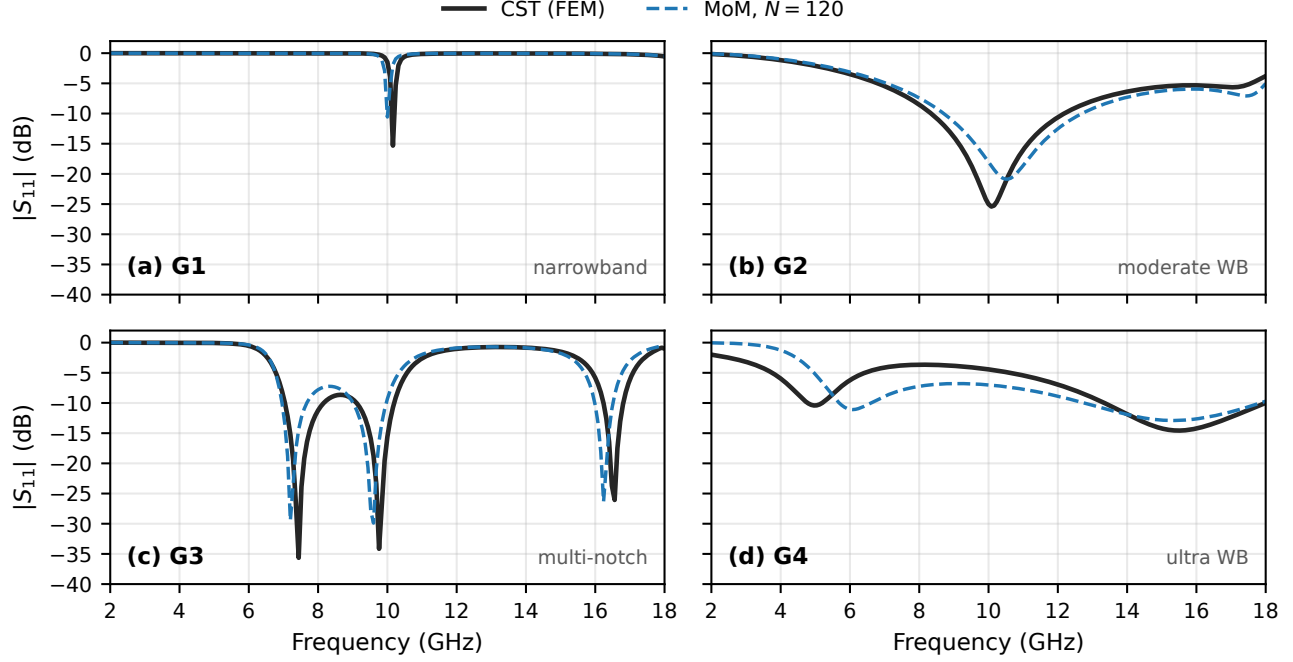


Fig. 3. Comparison of $|S_{11}|$ (dB) computed by the proposed solver ($N = 120$) and by the commercial FEM solver for (a) G1, (b) G2, (c) G3 and (d) G4.

TABLE IV
RESONANT NOTCHES: FEM (CST) VS PROPOSED SOLVER ($N = 120$)

Geom.	Notch	f_{notch} (GHz)		Δf (%)	Depth (dB)	
		FEM	MoM		FEM	MoM
G1	1	10.16	10.00	-1.57	-15.31	-10.58
G2	1	10.08	10.56	+4.76	-25.41	-20.87
G3	1	7.44	7.20	-3.23	-35.65	-29.46
G3	2	9.76	9.60	-1.64	-34.15	-29.90
G3	3	16.56	16.24	-1.93	-26.06	-26.32
G4	1	15.52	15.36	-1.03	-14.57	-12.90

$\Delta f = (f_{\text{MoM}} - f_{\text{FEM}})/f_{\text{FEM}}$. G4 also has a weaker first dip (FEM 4.96 GHz, MoM 6.08 GHz).

TABLE V
RUNTIME FOR ONE 201-POINT SWEEP ($N = 120$, 2–18 GHz)

	CST (s)	CPU (s)	A100 (s)	A100 speedup	
				vs CPU	vs CST
G1	182	24	0.72	33×	253×
G2	199	41	1.06	39×	188×
G3	117	44	1.07	41×	109×
G4	29	61	1.23	50×	24×

HPC NVIDIA A100. CPU: same FFT-MoM, NumPy/SciPy, complex128.

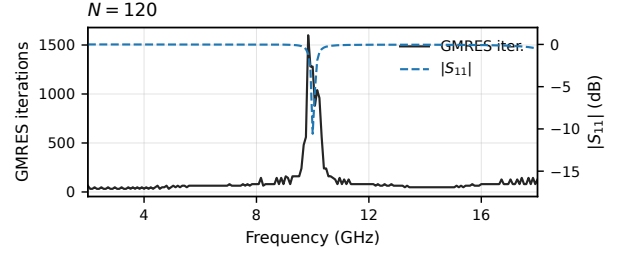


Fig. 4. GMRES iteration count across the $N = 120$ sweep of G1, with $|S_{11}|$ on the right axis.

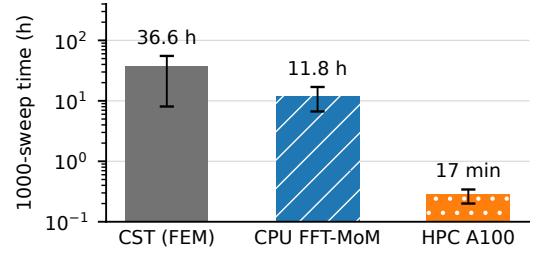


Fig. 5. Projected wall-clock time for 1000 sweeps (201 points each, $N = 120$, 2–18 GHz), taken as 1000 times the measured single-sweep time. Bars show the mean over G1–G4; whiskers span the fastest and slowest geometry.

V. CONCLUSION

A spectral-domain MoM solver for metasurface absorbers on grounded substrates has been presented. Folding the slab and ground into closed-form TE/TM Floquet Green's functions and applying the operator with FFTs avoids a dense impedance matrix; GMRES on the GPU reduces a full sweep by up to $50\times$ relative to the same algorithm on the CPU. Principal notches agree with commercial FEM to within 4.8% in frequency (depths within 6.2 dB). The residual error has not yet been separated by cause; likely contributors are the staircase approximation of curved edges, Floquet aliasing, and the zero-thickness sheet model. Future work includes oblique incidence, multilayer stacks, and training-data generation.

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